Lab 1 - DocStock Product Description Outline

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CS 410

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13 October 2026

Final Version

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1. Introduction

Managing medication is a significant challenge for many people. Remembering when to take each medication, following dosage instructions, tracking if/when the medication has been taken, and maintaining a proper supply of the medications can become difficult as the number of medications increase. Medication nonadherence is associated with not only poorer health outcomes, but also an increase in health costs. According to a 2017 Centers for Disease Control and Prevention (CDC) report, about one in five new prescriptions in the United States are never filled, while approximately 50% of filled prescriptions are taken incorrectly with regards to timing, dosage, frequency, and duration. The CDC has reported that direct health care costs associated with nonadherence have grown to approximately $100-$300 billion of U.S. health care dollars spent annually [1].

These challenges demonstrate that there is a need for a medicine-management system. One that is accessible, simply organized, and easy to use. An effective solution should help users maintain their medication schedule, provide reminders when it is time to medicate, track their dosages, allow users to acknowledge when they have taken their medications, track their current medication supply, and identify potential drug interactions. It should also support caregivers to do the same. DocStock is our proposed solution. DocStock will be a web and mobile application designed to address all these needs. The application is designed to allow users to set up their schedules, input and update medications, track dosages and when they've taken their medicine, and notify the user when they need a refill. DocStock will also alert users when there may be an unwanted drug interaction. Through prototyping, DocStock will demonstrate and the team will evaluate these features and how effective they may be. The prototype will provide a reasonable representation of DocStock's intended functionality and will demonstrate all our major features through simulated users.

2. DocStock Product Description

DocStock is a web and mobile app that helps individuals and their caregivers manage their prescriptions by scheduling doses, screening drugs for interactions, and tracking medication supply, with the overall goal of increasing medication adherence. DocStock's number one priority is simplifying what can be a stressful process into something easy to understand and manage. Our solution is not only to have DocStock notify users when it is time to take their medications, but also to make approved caregivers aware of when and what their dependents have missed. We will do so by allowing users to connect a caregiver to their account, giving that caregiver access to important medication information and missed-dose notifications. DocStock will also keep track of how much medication a user has left and alert them when their supply is running low. By combining reminders, caregiver support, interaction screening, and supply tracking into one

application, our objective is to give users a simple and organized way to stay on top of their medications and reduce the chances of missed, late, or repeated doses.

2.1. Key Product Features and Capabilities

By utilizing Dockstock, tracking prescriptions will be easier for individuals as all their prescriptions will be centralized into one easy to navigate dashboard that is reachable from any device where adherence is tracked and medications are listed. The app also provides a scheduling feature which allows users to schedule reminders for their doses and alert users when they have an upcoming refill. But what makes Dockstock different from any other reminders app? Dockstock has multiple innovative features, such as screening drugs for interactions, generating reports, and providing users with a caregiver mode. After every new prescription is added to a user's active list, Dockstock screens for any drug interactions, warning users if there are any possible effects. Users can also access daily, monthly, or yearly reports that track their prescription use, refills, and medications and generate these reports into an organized file to print or send to appointments. Dockstock's most unique feature is its caregiver mode, which allows users to create a caregiver profile that can assign reminders and schedule doses to their patient/client. These features make it easier for individuals and caregivers to manage multiple medication prescriptions conveniently and effectively by offering specialized support that other traditional reminders apps may not be able to provide.

2.2. Major Components (Hardware/Software)

Docstock will require Internet-connected devices to support development and use of the system. The system is structured according to the Major Functional Components Diagram. Users will interact with Docstock through a computer or smartphone. The web application will utilize HTML, CSS, and JavaScript and the mobile application is developed through React Native. The applications will provide features including account management, medication management, notifications, tracking, and caregiver support. The frontend applications will communicate with the backend using Node.js and Express. The backend handles scheduling, notifications, and business logic. The backend will also communicate between applications and the database. The database will store user accounts, medication information, history, and supply data through MySQL. For software, Visual Studio Code will be used as the development environment for writing and editing code. GitHub and GitHub Actions will support version control, automated testing, and continuous integration. The React Native Testing Library will be used

to test the mobile application's functionality. These software components will provide functionality for the features included in the components diagram.

3. Identification of Case Study

For whom is this product being developed? Why? Who else might use this in the future?

4. Glossary

4.1. Adherence: The extent to which a user follows their prescribed medication schedule, including taking medication at the correct time, frequency, and dosage.

4.2. Drug Interaction: A potentially unwanted effect that can occur when two or more medications interact with each other.

4.3. Nonadherence: Failure to correctly follow a prescribed medication regimen.

4.4. Prescription: A medication ordered for a patient by a licensed provider.

5. References

[1] Centers for Disease Control and Prevention, "CDC Grand Rounds: Improving medication adherence for chronic disease management — Innovations and opportunities," Morbidity and Mortality Weekly Report, vol. 66, no. 45, pp. 1248–1251, Nov. 2017. [Online]. Available: https://www.cdc.gov/mmwr/volumes/66/wr/mm6645a2.htm. [Accessed: Aug. 29, 2026].

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